

## DIFFRACTION PROBLEMS

**Q1.** A diffraction grating used at normal incidence gives a yellow line ( $\lambda = 6000 \text{ \AA}$ ) in a certain spectral order superimposed on a blue line ( $\lambda = 4800 \text{ \AA}$ ) of next higher order. If the angle of diffraction is  $\sin^{-1}(3/4)$  calculate the grating element.

**Given:-**  $\lambda_1 = 6000 \text{ \AA} = 6 \times 10^{-5} \text{ cm}$ ;  $\lambda_2 = 4800 \text{ \AA} = 4.8 \times 10^{-5} \text{ cm}$ ;  $\theta = \sin^{-1}\left(\frac{3}{4}\right)$

**Formula:-**  $(a + b)\sin \theta = n \lambda$ ;  $n = 1, 2, 3, 4, \dots$

**Solution:-** for given  $(a + b)$  and  $\theta$ ;  $n \propto 1/\lambda$

$$(a + b)\sin \theta = n \lambda_1$$

$$(a + b)\sin \theta = (n + 1) \lambda_2$$

$$n \lambda_1 = (n + 1) \lambda_2$$

$$\frac{\lambda_1}{\lambda_2} = 1 + \frac{1}{n}$$

Therefore,  $n = 4$

$$(a + b) = \frac{n \lambda_1}{\sin \theta} = \frac{4 \times 6 \times 10^{-5}}{\frac{3}{4}} = 32 \times 10^{-5} \text{ cm}$$

**Ans:- The grating element is  $3.2 \times 10^{-4} \text{ cm}$ .**

**Q2.** Monochromatic light of wavelength  $6560 \text{ \AA}$  falls normally on a grating  $2 \text{ cm}$  wide. The first order spectrum is produced at an angle of  $16^\circ 17'$  from the normal. Calculate the total number of lines on the grating.

**Given:-**  $\lambda = 6560 \text{ \AA} = 6560 \times 10^{-8} \text{ cm}$ ; width =  $2 \text{ cm}$ ;  $n = 1$ ;  $\theta = 16.28^\circ$

**Formula:-**  $(a + b)\sin \theta = n \lambda$

$$a + b = \frac{1}{(N) \text{ Number of lines per cm}}$$

Total number of lines =  $N \times \text{width}$

**Solution:-**  $(a+b) = \frac{n\lambda}{\sin\theta} = \frac{6560}{\sin 16.28} \times 10^{-8} = 2.34 \times 10^{-4} \text{ cm}$

$$\text{Number of lines per cm} = \frac{1}{a+b} = 4273$$

$$\text{Total no. of lines} = 4273 \times 2 = 8547$$

**Ans:- There will be 8547 lines on the grating.**

**Q3.** A parallel beam of light is incident on a plane transmission grating having 3000 lines/cm. A third order diffraction is observed at  $30^\circ$ . calculate the wavelength of the line.

**Given:-**  $a+b = 1/3000$  ;  $n=3$  ;  $\theta=30$

**Formula:-**  $(a + b)\sin\theta = n\lambda$  ,  $n=1,2,3...$

**Solution:-**  $\lambda = \frac{a+b}{n} \times \sin\theta$

$$= \frac{1}{3000 \times 3} \times \sin 30$$
$$= \frac{1}{9000 \times 2} = 5.555 \times 10^{-5} \text{ cm}$$

**Ans:- The wavelength of the line is  $5.555 \times 10^{-5} \text{ cm}$ .**

**Q4.** The visible spectrum ranges from  $4000\text{\AA}$  to  $7000\text{\AA}$ . Find the angular breadth of the first order visible spectrum produced by a plane grating having 6000 lines/cm when light is incident normally on the grating.

**Given:-**  $l_1 = 4000\text{\AA} = 4 \times 10^{-5} \text{ cm}$      $l_2 = 7000\text{\AA} = 7 \times 10^{-5} \text{ cm}$      $n=1$   
 $a+b = 1/6000 \text{ lines per cm}$

**Formula:-**  $(a + b)\sin\theta = n\lambda$

**Solution:-**  $(a + b)\sin\theta_1 = \lambda_1$

$$\theta_1 = \sin^{-1} \frac{\lambda_1}{a+b} = \sin^{-1}(4 \times 10^{-5} \times 6000) = 13.88^\circ$$

$$(a+b) \sin \theta_2 = \lambda_2$$

$$\theta_2 = \sin^{-1} \frac{\lambda_2}{a+b} = \sin^{-1}(7 \times 10^{-5} \times 6000) = 24.83^\circ$$

$$\theta_2 - \theta_1 = 24.83 - 13.88 = 10.95^\circ$$

**Ans :- The Angular separation = 10.95°**

**Q5.** In plane transmission grating the angle of diffraction for the second order principal maxima for the wavelength  $5 \times 10^{-5}$  cm is  $35^\circ$ . Calculate the number of lines/cm on the diffraction grating.

**Given:-**  $\lambda = 5 \times 10^{-5}$  cm ;  $\theta = 35^\circ$  ;  $n = 2$

**Formula:-**  $(a+b) \sin \theta = n \lambda$ ;  $\frac{1}{a+b}$  = number of lines/cm

**Solution:-**  $a+b = \frac{n\lambda}{\sin \theta} = \frac{2 \times 5 \times 10^{-5}}{\sin 35^\circ} = 1.74 \times 10^{-4}$

$$\text{Number of lines per cm} = \frac{1}{a+b} = \frac{1}{1.74 \times 10^{-4}} = 5735$$

**Ans:- The number of lines per cm is 5735.**

**Q6.** A grating has 620 ruling/mm and is 0.5 mm wide. What is the smallest wavelength interval that can be resolved in the third order at  $\lambda = 481 \text{ nm}$ ?

**Given:-**  $N = 620 \times 0.5 = 310$  ;  $\lambda = 481 \times 10^{-9} \text{ m}$  ;  $m = 3$

**Formula:-**  $\frac{\lambda}{d\lambda} = mN$

**Solution:-**  $d\lambda = \frac{\lambda}{mN} = \frac{481 \times 10^{-9}}{3 \times 310} = 0.5172 \times 10^{-9} \text{ m}$

$$d\lambda = 0.5172 \text{ Å}$$

**Ans:- The smallest wavelength interval is 0.5172 Å**

**Q7.** Calculate the minimum number of lines required on grating that can just resolve the two sodium lines  $\lambda_1=5890\text{A}$  and  $\lambda_2=5893\text{ A}$  in third order.

**Given:-**  $\lambda_1=5890 \times 10^{-8}\text{ cm}$ ,  $\lambda_2=5893 \times 10^{-8}\text{ cm}$ ,  $m=3$

**Formula:-** Resolving power  $= \frac{\lambda}{d\lambda} = mN$

**Solution:-**  $\lambda = \frac{\lambda_1 + \lambda_2}{2} = \frac{(5890+5893)10^{-8}}{2} = 5893 \times 10^{-8}\text{ cm}$

$$d = (5893-5890) \times 10^{-8} = 3 \times 10^{-8}\text{ cm}$$

$$N = \frac{\lambda}{m d \lambda} = \frac{5893 \times 10^{-8}}{3 \times 3 \times 10^{-8}} = 327$$

**Ans:- Minimum of 327 lines are required on the grating.**

**Q8.** Calculate the maximum order of diffraction maxima seen from plane transmission grating with 2500 lines per inch if light of wavelength 6900 A falls normally on it.

**Given:-**  $N = \frac{1}{a+b} = 2500\text{ lines/inch} = 2500 \times 2.52 \times 10^{-2} = 63\text{ lines/m}$

$$\lambda = 6900\text{A} = 6900 \times 10^{-10}\text{ m}$$

**Formula:-**  $(a + b)\sin \theta = n \lambda$

**Solution:-** for  $n = n_{\max}$ ,  $\sin \theta = 1$

$$n_{\max} = \frac{a+b}{\lambda} = 2.3$$

**Ans:- Maximum order of diffraction is 2**